

Tutorial - 4

1. The switch in **Fig. 2** was open for a long time but closed at $t = 0$.

Determine: (a) $i(0^+)$, $v(0^+)$, (b) $di(0^+)/dt$, $dv(0^+)/dt$, (c) $i(\infty)$, $v(\infty)$.

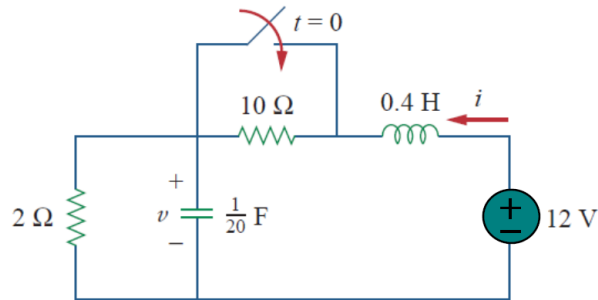
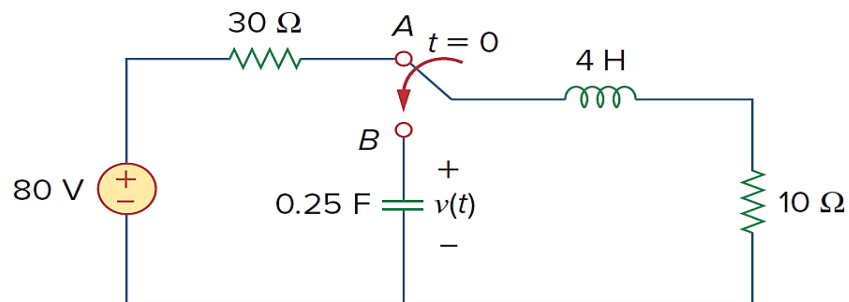


Fig. 2

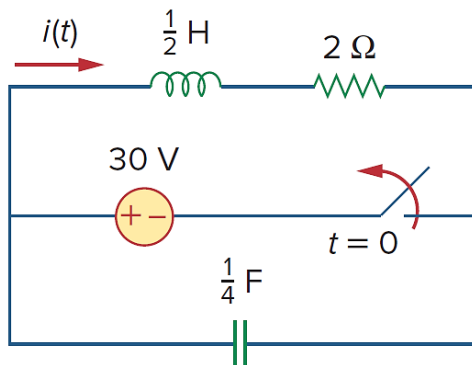
2. The switch moves from position A to position B at $t = 0$. Let $v(0) = 0$, find $v(t)$ for $t > 0$.



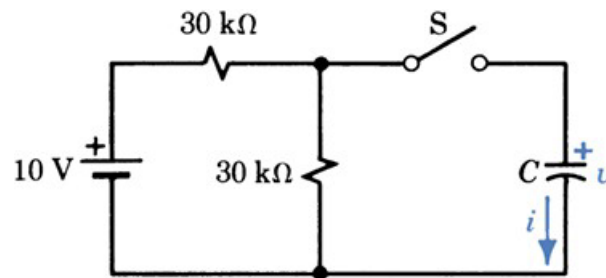
3. A source-free RLC circuit has $R = 1 \Omega$, $C = 1 \text{ nF}$, and $L = 1 \text{ nH}$. Calculate the following:

- Calculate α and ω_0 .
- Calculate s_1 and s_2 .
- What is the form of the inductor current response for $t > 0$?

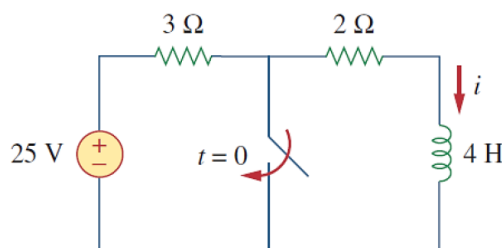
4. The switch in the circuit has been closed for a long time but is opened at $t = 0$. Determine $i(t)$ for $t > 0$.



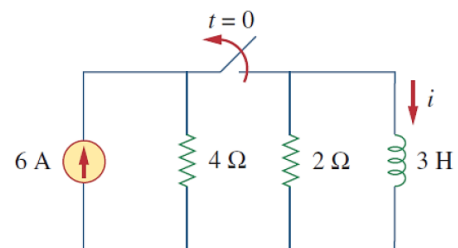
5. Switch S is closed at $t = 0$. Calculate voltage v and the current i when $C = 3 \mu F$.



6. Obtain the inductor current ' $i(t)$ ' for $t > 0$ in each of the circuits.

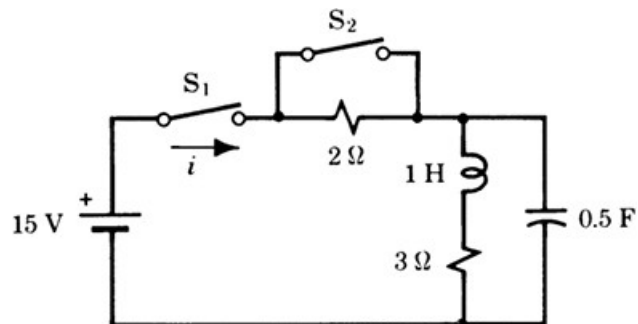


(a)



(b)

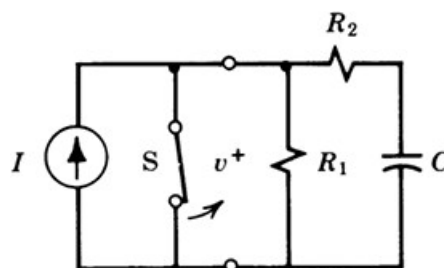
7. Switches S_1 and S_2 have been closed for a long time.



- Where in this circuit is energy stored and how much?
- If S_2 is opened at $t = 0$, calculate i_L , v_C , i , and i_C and $t = 0^+$
- Derive an expression for $i(t)$.

8. The switch S is opened at $t = 0$. Determine:

- The steady state response component of v .
- The natural response component of v .
- The complete response $v(t)$.



9. TBD: check after some time
10. TBD: check after some time
11. TBD: check after some time